

Activity 10 Pre-Lab — Crossed Aldol (2-Acetylpyridine + 4-Nitrobenzaldehyde)

[Your name] & [partner]
[MM/DD/YY]

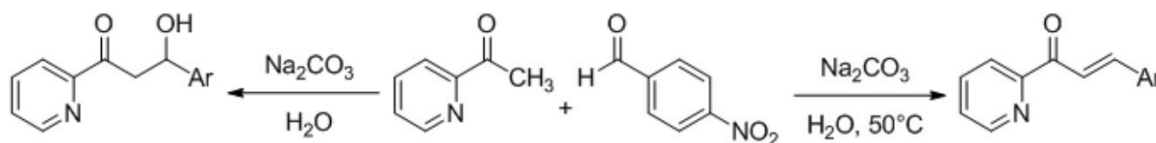
Organic Lab · aldol addition vs. condensation

■ Black = write this in your notebook ■ Blue = context, don't copy it Copy the header onto every page. Draw dashed-box structures yourself.

Purpose

Run a base-catalyzed **crossed aldol** of **2-acetylpyridine** (the enolate) with **4-nitrobenzaldehyde** (the electrophile), and show how **reaction conditions** decide the product: **room temp** → **aldol addition** (β -hydroxy ketone) vs **heat (50 °C)** → **aldol condensation** (the α,β -unsaturated ketone + H_2O).

🔥 DRAW — reproduce this in your notebook (from the procedure)



base (Na_2CO_3) makes the enolate at the acetyl CH_3 → it attacks the aldehyde. **RT** stops at the β -hydroxy ketone (addition, left); **50 °C** dehydrates to the α,β -unsaturated ketone (condensation, right) + H_2O .

Chemical Reagents & Safety Table

CONTEXT ▶ Required: 2-acetylpyridine, 4-nitrobenzaldehyde, sodium carbonate, methanol — plus a row for each product (MW only, per directions).
Typical SDS values — cross-check the SDS tile.

Chemical	Role	MW	Amount	Physical properties	Hazards (GHS)
2-Acetylpyridine	Enolate (nucleophile)	121.14	~0.2 mL (weigh exact)	Yellow liquid; bp $192\text{ }^\circ\text{C}$; d 1.08 g/mL	H302 harmful if swallowed · H315 · H319 · H335
4-Nitrobenzaldehyde	Aldehyde (electrophile)	151.12	300 mg	Pale-yellow solid; mp $106\text{ }^\circ\text{C}$	H315 skin irritation · H319 eye irritation · H335 respiratory irritation
Sodium carbonate (0.54% w/v)	Base (makes enolate)	105.99	10 mL	Colorless aqueous solution	H319 eye irritation
Methanol	Dissolves the aldehyde	32.04	5–7 mL	Colorless liquid; bp $65\text{ }^\circ\text{C}$; d 0.79 g/mL ; flash pt $11\text{ }^\circ\text{C}$	H225 highly flammable · H301+H311+H331 toxic · H370 organ damage (eyes/CNS)
Addition product (β -hydroxy ketone)	Product — Part 1 (RT)	272.26	— (MW only)	—	—
Condensation product (enone)	Product — Part 2 (50 °C)	254.24	— (MW only)	—	—

FYI ▶ condensation MW = addition MW – 18 (loss of one H_2O). Each pair runs Part 1 or Part 2, then shares data.

GRAB FIRST: 100-mL round-bottom + stir bar · 1-mL syringe · cork ring (for weighing) · 25-mL Erlenmeyer · steam bath · funnel · stir plate on a raised lab jack (Part 2 also: heat block + condenser to 50 °C) · Büchner + filter flask + vacuum.

Procedure Plan (left = plan, right = fill in during lab)

Plan (write before lab)

Reaction (Part 1 = RT · Part 2 = 50 °C)

1. Tare a cork ring; put a stir bar + 1-mL syringe in a **100-mL RBF**; weigh the assembly (record).
2. Draw **~0.2 mL 2-acetylpyridine** into the syringe at the dispensing station; reweigh assembly → **exact mass of 2-acetylpyridine**.
3. Add the 2-acetylpyridine into the RBF + **20 mL H₂O**. *Rinse syringe w/ acetone into waste, discard.*
4. In a 25-mL Erlenmeyer dissolve **300 mg 4-nitrobenzaldehyde in 5 mL MeOH** over a steam bath (add 2 mL more MeOH if needed). While warm, add it to the RBF via funnel.
5. Add **10 mL 0.54% Na₂CO₃** through the same funnel (base = starts the enolate).
6. Stir vigorously **60 min**. *Part 1: room temp → stops at the addition product. Part 2: 50 °C (heat block + condenser, watch for flooding) → drives dehydration to the condensation product. Keep organic solvents away from the hot block.*
7. Collect solid by **vacuum filtration**; wash **2 × 20 mL DI water**; pull vacuum 10 min; scrape onto filter paper & label (ADDITION or CONDENSATION).
8. Weigh → % yield; take mp / IR / NMR; compare with the other group's product.

WRITE

limiting reagent (compare mol 2-acetylpyridine vs 4-nitrobenzaldehyde 300 mg ÷ 151.12) → theoretical product → % yield.

Experiment Notes (in lab)

Which part (RT / 50 °C): _____

Mass assembly: _____ g after 2-acetylpyridine: _____ g

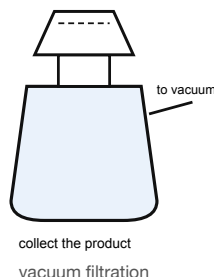
Mass 2-acetylpyridine: _____ g

Appearance during stir (solid forming? color):

Product mass / mp: _____

% yield · addition vs condensation compare:

Setup — draw this



CONTEXT (don't copy) ▶

Addition vs condensation: the enolate adds to the aldehyde to give a β-hydroxy ketone (the *addition* product). Heating (and the electron-poor 4-nitro ring) makes it lose water (E1cb) to give the conjugated **α,β-unsaturated ketone** (the *condensation* product). Same starting materials — the temperature decides.

Why it precipitates: the conjugated nitro product is a bright, insoluble solid → easy vacuum filtration.

Clean-up & Conclusions

- Filtrates → aqueous waste; MeOH/organic → organic waste; syringe rinses → labeled waste; return glassware; bench task.
- **Conclusions** (~4 sentences *after lab*): which product your conditions gave and why (RT vs 50 °C); % yield; how mp/IR/NMR distinguish addition (has O-H, sp³ C-H) from condensation (C=C, conjugation); comparison with the other group.